

Windows Timeline and Timestomping Analysis

Professional Case Report 02

Author	Balsa Klikovac
Report date	24. septembar 2026
Case ID	CSC-2026-02
Discipline	Digital forensics
Overall risk	Critical
Evidence status	Confirmed by saved Redline evidence

Purpose

Build an attack timeline, identify dual-use tools and distinguish displayed file times from true NTFS creation evidence.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

The initial lead was expanded into a ten-minute forensic window around 02:57:08 UTC to identify additional attacker activity and possible timestamp manipulation. The evidence supports a coordinated post-compromise sequence involving credential access, SYSTEM privileges, remote administration tools and timestamp manipulation.

Case Profile

Scenario	The initial lead was expanded into a ten-minute forensic window around 02:57:08 UTC to identify additional attacker activity and possible timestamp manipulation.
Environment	Windows endpoint collection analyzed in Redline with raw MFT timestamps.
Tools	Mandiant Redline, MFT timeline analysis, VirusTotal research, File signature validation

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Applied a creation-time filter from 02:52:08 to 03:02:08 UTC.
2. Excluded high-volume Windows paths to focus on non-standard directories.
3. Reviewed all files in C:\tmp and compared SI and FN timestamps.
4. Classified tools by capability and reconstructed the sequence of activity.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	mimi.zip	Critical	MD5 c7848c3dd0aec5055bc6983b0cf89310; archive associated with credential extraction tooling.
F02	Psexec.exe	High	Signed Microsoft Sysinternals utility with MD5 27304b246c7d5b4e149124d5f93c5b01; legitimate administration tool commonly abused for remote execution.
F03	wfreerdp.exe and z.exe	High	Unsigned executables had implausible 1940 timestamps, indicating timestamping or metadata manipulation.
F04	whoami.tct	High	Output indicated NT AUTHORITY\SYSTEM context, supporting successful privilege escalation.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

Redline time filter: 2017-06-13 02:52:08 to 03:02:08 UTC

Compare SI Created with Filename Created timestamps

Risk Assessment

Risk	Level	Operational impact
Credential theft	Critical	Loss of privileged credentials and domain compromise
Lateral movement	High	Remote execution through PsExec and RDP tooling
Anti-forensics	High	Timestomping weakens simple timeline review

Recommendations

1. Reset credentials exposed on the endpoint and invalidate active sessions.
2. Search enterprise telemetry for PsExec, FreeRDP and matching hashes.
3. Use MFT Filename timestamps when SI values are implausible.
4. Restrict remote administration tools through application control.

Knowledge Demonstrated

- Timeline reconstruction
- Timestomping detection
- Dual-use tool analysis
- Privilege assessment

Conclusion

The evidence supports a coordinated post-compromise sequence involving credential access, SYSTEM privileges, remote administration tools and timestamp manipulation.

Hands On Lab Practical 02

Completed as part of the authorized cyber security course practical work.